

Almost all half-degree patterns are realized by an irreducible over $\mathbb{F}_2$

Michael Bommarito

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Abstract

Write $\ell = \lceil n/2 \rceil - 1$. For each pattern $h \in \mathbb{F}_2^\ell$ of the top ℓ non-leading coefficients of a monic degree- n polynomial over $\mathbb{F}_2$, we ask whether some irreducible of degree n carries that pattern. We prove, unconditionally, that all but at most $4\ell^2$ of the 2^ℓ patterns are realized (Theorem B), via a sharp Parseval–Weil second-moment bound on the ray class group (Theorem A). The Kaser–Lemire conjecture is exactly the assertion that *one specific* pattern – the identity class $h = 0$ – is not among the exceptions, and we show why a second-moment argument cannot decide any named class: there is an explicit nonnegative function on the class group with the same low-order Fourier data as the true count that vanishes at the identity (Proposition 1). This companion to the proof roadmap [1] records what the analytic input proves on its own and marks precisely where it runs out.

1 Setting and the three inputs

Fix n and put $\ell = \lceil n/2 \rceil - 1$; we treat the two endpoints $n \in \{2\ell + 1, 2\ell + 2\}$ that the conjecture concerns. Let

$$\mathcal{E}_\ell = (1 + x\mathbb{F}_2[x])/(x^{\ell+1}), \quad |\mathcal{E}_\ell| = 2^\ell,$$

be the ray class group, and for monic F write $\langle F \rangle_\ell = x^{\deg F} F(x^{-1}) \bmod x^{\ell+1}$ for its reversed class. For $\deg F = n$ this reads $\langle F \rangle_\ell = 1 + a_{n-1}x + \cdots + a_{n-\ell}x^\ell$, so a class $g \in \mathcal{E}_\ell$ is exactly a *top-half pattern* $h = (a_{n-1}, \dots, a_{n-\ell}) \in \mathbb{F}_2^\ell$. Weight each class by the polynomial von Mangoldt function Λ :

$$N_\ell(g) = \sum_{\substack{\deg F = n \\ \langle F \rangle_\ell = g}} \Lambda(F), \quad S_n(\chi) = \sum_{\deg F = n} \Lambda(F) \chi(\langle F \rangle_\ell) \quad (\chi \in \widehat{\mathcal{E}_\ell}).$$

The total mass is $\sum_g N_\ell(g) = S_n(\mathbf{1}) = 2^n$, so the mean of N_ℓ over the 2^ℓ classes is $2^{n-\ell}$. Our arguments use only three inputs about these quantities.

(O) Orthogonality. Fourier inversion on $\mathcal{E}_\ell$ gives $N_\ell(g) = 2^{-\ell} \sum_\chi S_n(\chi) \overline{\chi(g)}$, and Parseval gives $\sum_g (N_\ell(g) - 2^{n-\ell})^2 = 2^{-\ell} \sum_{\chi \neq \mathbf{1}} |S_n(\chi)|^2$.

(W) The Weil bound. If χ has conductor j then $L(u, \chi) = \prod_{i \leq j-1} (1 - \alpha_i u)$ with $|\alpha_i| \in \{0, \sqrt{2}\}$ and $S_n(\chi) = -\sum_i \alpha_i^n$, so $|S_n(\chi)| \leq (j-1)2^{n/2} \leq (j-1)2^{\lceil n/2 \rceil}$. The trivial conductor $j = 1$ gives $L = 1$, hence $S_n = 0$ there.

(C) The conductor count. Exactly 2^{j-1} characters have conductor equal to j for each $1 \leq j \leq \ell$ (plus the trivial one). None of the three distinguishes one class of $\mathcal{E}_\ell$ from another: (O), (W), (C) are invariant under every permutation of the group – harmless for the average statements of §2–§3, fatal for the named class of §4. We will use the following closed form; write $A(\ell) = \sum_{i=1}^{\ell-1} i^2 2^i$.

Lemma 1. *For every integer $m \geq 0$, $\sum_{i=1}^m i^2 2^i = 2^{m+1}(m^2 - 2m + 3) - 6$. Consequently $A(\ell) = 2^\ell(\ell^2 - 4\ell + 6) - 6$.*

Proof. Induct on m . For $m = 0$ both sides are 0. Assuming the formula at m , add the next term $(m+1)^2 2^{m+1}$:

$$\sum_{i=1}^{m+1} i^2 2^i = 2^{m+1}(m^2 - 2m + 3) - 6 + 2^{m+1}(m^2 + 2m + 1) = 2^{m+1}(2m^2 + 4) - 6 = 2^{m+2}(m^2 + 2) - 6.$$

This is the target at $m+1$, since $(m+1)^2 - 2(m+1) + 3 = m^2 + 2$. The specialization $m = \ell - 1$ gives $A(\ell) = 2^\ell((\ell-1)^2 - 2(\ell-1) + 3) - 6 = 2^\ell(\ell^2 - 4\ell + 6) - 6$. $\square$

2 Theorem A: almost all classes carry mean mass

Put

$$\varepsilon(\ell) = (\ell^2 - 4\ell + 6) - 6 \cdot 2^{-\ell} = A(\ell) 2^{-\ell}, \quad \kappa_n = 2^{2\lceil n/2 \rceil - n} \in \{1, 2\}.$$

Theorem A. Let $\ell \geq 1$, $n \in \{2\ell + 1, 2\ell + 2\}$, and $0 < t \leq 1$. Then

$$\#\{g \in \mathcal{E}_\ell : |N_\ell(g) - 2^{n-\ell}| \geq t 2^{n-\ell}\} \leq \frac{\kappa_n 2^{2\ell-n} \varepsilon(\ell)}{t^2} = \begin{cases} \varepsilon(\ell)/t^2, & n = 2\ell + 1, \\ \varepsilon(\ell)/(4t^2), & n = 2\ell + 2, \end{cases}$$

a fraction at most $\varepsilon(\ell)/(t^2 2^\ell)$ of the 2^ℓ classes.

Proof. Write $D(g) = N_\ell(g) - 2^{n-\ell}$. By (O), $\sum_g D(g)^2 = 2^{-\ell} \sum_{\chi \neq 1} |S_n(\chi)|^2$. Group the nontrivial characters by conductor and apply (W) and (C):

$$\sum_{\chi \neq 1} |S_n(\chi)|^2 \leq 2^{2\lceil n/2 \rceil} \sum_{j=1}^{\ell} 2^{j-1} (j-1)^2 = 2^{2\lceil n/2 \rceil} \sum_{i=1}^{\ell-1} i^2 2^i = 2^{2\lceil n/2 \rceil} A(\ell),$$

where the second equality reindexes $i = j - 1$ (the $j = 1$ term vanishes, consistent with $S_n = 0$ at conductor 1). By Lemma 1, $A(\ell) = 2^\ell \varepsilon(\ell)$, so

$$\sum_g D(g)^2 \leq 2^{-\ell} \cdot 2^{2\lceil n/2 \rceil} \cdot 2^\ell \varepsilon(\ell) = 2^{2\lceil n/2 \rceil} \varepsilon(\ell) = \kappa_n 2^n \varepsilon(\ell).$$

Chebyshev's inequality at threshold $t 2^{n-\ell}$ gives

$$\#\{g : |D(g)| \geq t 2^{n-\ell}\} \leq \frac{\sum_g D(g)^2}{t^2 2^{2(n-\ell)}} \leq \frac{2^{2\lceil n/2 \rceil} \varepsilon(\ell)}{t^2 2^{2(n-\ell)}} = \frac{\kappa_n 2^{2\ell-n} \varepsilon(\ell)}{t^2}.$$

For $n = 2\ell + 1$ one has $\lceil n/2 \rceil = \ell + 1$, so $\kappa_n = 2$ and $\kappa_n 2^{2\ell-n} = 1$; for $n = 2\ell + 2$ one has $\kappa_n = 1$ and $\kappa_n 2^{2\ell-n} = \frac{1}{4}$. Dividing by 2^ℓ gives the stated fraction. $\square$

Using the sharper $|S_n(\chi)| \leq (j-1)2^{n/2}$ sets $\kappa_n = 1$, halving the odd- n constant. The Weil input is provably lossy here by a factor of order ℓ : the Sato–Tate heuristic predicts $\sum_{\text{cond } j} |S_n|^2 \approx 2^{j-1} (j-1) 2^n$, replacing $A(\ell)$ by $2^\ell(\ell-2) + 2$. But that is a $q \rightarrow \infty$ statement while $q = 2$ is fixed, and by (O) the exact second moment *is* what we are bounding, so Parseval cannot supply it. The numerics of §4 sit at $1.00\times$ the Sato–Tate value, so the true saving is exactly this $\sim \ell$.

3 Theorem B: almost all patterns are realized

For a pattern $h \in \mathbb{F}_2^\ell$ let $I_n(h)$ be the number of monic irreducibles of degree n whose coefficients of $x^{n-1}, \dots, x^{n-\ell}$ equal h , equivalently the irreducibles in the class $g \leftrightarrow h$. Let $\Theta(g)$ be the Λ -mass of *proper* prime powers in class g , so $n I_n(g) = N_\ell(g) - \Theta(g)$, and globally $\Theta_n := \sum_g \Theta(g) = \sum_{d|n, d < n} d I_d$.

Theorem B. Let $\ell \geq 5$ and $n \in \{2\ell + 1, 2\ell + 2\}$. The number of patterns $h \in \mathbb{F}_2^\ell$ realized by no irreducible of degree n is at most

$$4(\ell^2 - 4\ell + 6) < 4\ell^2 \quad (n = 2\ell + 1), \quad (\ell^2 - 4\ell + 6) < \ell^2 \quad (n = 2\ell + 2).$$

Hence all but a fraction $< 4\ell^2 2^{-\ell}$ of the 2^ℓ top-half patterns are the top half of some irreducible of degree n – a positive proportion once $\ell \geq 9$.

Proof. We first bound the proper-power mass classwise. *Odd n .* Every proper prime power is P^k with k odd and $k \geq 3$, so its underlying prime has degree $d = n/k \leq n/3$; thus $\Theta_n \leq \sum_{d \leq n/3} 2^d < 2^{n/3+1}$. By Markov's inequality,

$$\#\{g : \Theta(g) \geq \tfrac{1}{2} 2^{n-\ell}\} \leq \frac{\Theta_n}{2^{n-\ell-1}} < 2^{\ell+2-2n/3} = 2^{(4-\ell)/3} < 1 \quad (n = 2\ell + 1, \ell \geq 5),$$

so this set is *empty*. *Even n .* Now $n/2$ is possible, and the squares of the degree- $n/2$ irreducibles contribute mass $\leq 2^{n/2}$, all higher powers $< 2^{n/3+1}$, so $\Theta_n \leq 2^{n/2} + 2^{n/3+1}$. Here $\frac{1}{2} 2^{n-\ell} = 2^{n-\ell-1} = 2^{n/2}$, and $\Theta_n/2^{n/2} \leq 1 + 2^{1-n/6} < 2$, so at most *one* class has $\Theta(g) \geq \frac{1}{2} 2^{n-\ell}$.

Now combine with Theorem A at $t = \frac{1}{2}$. Off that exceptional set one has $N_\ell(g) > 2^{n-\ell} - \frac{1}{2} 2^{n-\ell} = \frac{1}{2} 2^{n-\ell}$; off the Markov set one has $\Theta(g) < \frac{1}{2} 2^{n-\ell}$. On the intersection of the two complements, therefore,

$$n I_n(g) = N_\ell(g) - \Theta(g) > \tfrac{1}{2} 2^{n-\ell} - \tfrac{1}{2} 2^{n-\ell} = 0,$$

so $I_n(g) \geq 1$: the pattern is realized. Only the two exceptional sets can fail, and their sizes add. Theorem A at $t = \frac{1}{2}$ bounds its set by $4\kappa_n 2^{2\ell-n} \varepsilon(\ell)$, i.e. $\lfloor 4\varepsilon(\ell) \rfloor$ classes for odd n and $\lfloor \varepsilon(\ell) \rfloor$ for even n ; the Markov set adds 0 (odd) or 1 (even). Since $0 < 6 \cdot 2^{-\ell} < 1$ and $0 < 24 \cdot 2^{-\ell} < 1$ for $\ell \geq 5$, we have $\lfloor 4\varepsilon(\ell) \rfloor = 4(\ell^2 - 4\ell + 6) - 1 < 4(\ell^2 - 4\ell + 6)$ and $\lfloor \varepsilon(\ell) \rfloor + 1 = (\ell^2 - 4\ell + 6)$. These are the stated counts, and $4(\ell^2 - 4\ell + 6) < 4\ell^2$, $(\ell^2 - 4\ell + 6) < \ell^2$. $\square$

The same argument at $t = \frac{1}{4}$ gives $I_n(h) \geq 2^{n-\ell-1}/n$ – a large irreducible count – for all but $\leq 16\varepsilon(\ell) + 2$ patterns.

4 The residual: why this does not settle the conjecture

The Kaser–Lemire conjecture is the assertion $I_n(1) \geq 1$ for the identity class $g = 1$ (the pattern $h = 0$): that this *one named* element lies outside the exceptional set of Theorem B. The theorem counts exceptions but cannot name one, because its only inputs (O), (W), (C) are permutation-invariant, so anything they prove holds for every relabeling of $\mathcal{E}_\ell$ at once. The next result makes the obstruction concrete.

Proposition 1 (Second-moment barrier). *Let $a = \ell - \lceil \log_2 \ell \rceil - 1$ and $K = \ker(\mathcal{E}_\ell \rightarrow \mathcal{E}_{a-1})$, an elementary abelian group of order $2^{\ell-a+1}$. There is a function $F: \mathcal{E}_\ell \rightarrow \mathbb{R}$ with all of the following properties: $F \geq 0$; $\sum_g F(g) = 2^n$; $\widehat{F}(\chi) = \widehat{N}_\ell(\chi)$ for every χ of conductor $< a$; $|\widehat{F}(\chi)|$ lies within the Weil bound $(j-1)2^{\lceil n/2 \rceil}$ for every χ of conductor $j \geq a$; and $F(1) = 0$.*

Proof. Let m be the cylinder-mean population: $m(g) = 2^{a-1-\ell} N_{a-1}(\pi(g))$, where $\pi: \mathcal{E}_\ell \rightarrow \mathcal{E}_{a-1}$ is the projection and $2^{a-1-\ell} = |K|^{-1}$. Then m is constant on cosets of K , is nonnegative, has $\sum_g m = \sum_g N_{a-1} = 2^n$, and agrees with N_ℓ in every Fourier coefficient of conductor $\leq a-1$ while $\widehat{m}(\chi) = 0$ for conductor $\geq a$. Set

$$F = m + c(2^{a-1-\ell} \mathbf{1}_K - \delta_1), \quad c = \frac{m(1)}{1 - 2^{a-1-\ell}} > 0,$$

where $\mathbf{1}_K$ is the indicator of K and δ_1 of the identity. The perturbation has zero total mass ($2^{a-1-\ell}|K| = 1 = \sum \delta_1$), so $\sum F = 2^n$. At the identity, $F(1) = m(1) + c(2^{a-1-\ell} - 1) = m(1) - m(1) = 0$. Off K , $F = m \geq 0$; on K one has $m \equiv m(1)$ and δ_1 vanishes except at 1, so $F = m(1) + c2^{a-1-\ell} > 0$ there and $F(1) = 0$; thus $F \geq 0$. Finally the Fourier transform of $2^{a-1-\ell} \mathbf{1}_K - \delta_1$ equals $[\chi|_K = \mathbf{1}] - 1$, which is 0 at conductor $< a$ (there χ is trivial on K) and -1 at conductor $\geq a$. Hence $\widehat{F} = \widehat{N}_\ell$ below conductor a , while at conductor $j \geq a$ we get $|\widehat{F}(\chi)| = c$. Now $c \approx m(1) = 2^{a-1-\ell} N_{a-1}(1) \approx 2^{n-\ell} = 2^{\lceil n/2 \rceil}$ at the odd endpoint (and smaller at the even one), so $c \leq (j-1)2^{\lceil n/2 \rceil}$ for every $j \geq a \geq 2$ – indeed well within Weil, by a factor of order a . $\square$

Every hypothesis a Theorem-A-style argument can use about N_ℓ – nonnegativity, exact total mass, low-conductor Fourier data, and Weil-admissible moduli above – holds verbatim for F , yet $F(1) = 0$. So no argument built solely from mass, nonnegativity, Fourier moduli, and low moments can force $N_\ell(1) > 0$. Sharpening $\varepsilon(\ell)$, or even proving the Sato–Tate second moment, cannot break this barrier, since F already satisfies the stronger moment.

What *could* decide a named class falls into three kinds of input, none of them permutation-invariant.

- (1) **A transitive symmetry commuting with N_ℓ :** a group action on $\mathcal{E}_\ell$ commuting with the count whose orbit of the identity meets the non-exceptional set. None exists: every degree-preserving symmetry lies in $\mathrm{PGL}_2(\mathbb{F}_2)$, and only its Borel subgroup $\{\mathrm{id}, f(x) \mapsto f(x+1)\}$ descends to $\mathcal{E}_\ell$ commuting with N_ℓ . Translation is an involution carrying the identity class to $\langle (1+x)^n \rangle_\ell$, and the Galois/Adams action fixes it, so the orbit of the identity has size at most $2 < 4\ell^2$; the multiplication action is transitive on $\mathcal{E}_\ell$ but shifts the degree. Hence no symmetry helps.
- (2) **The class’s own description:** the identity class is the short interval $\{x^n + g : \deg g \leq \lfloor n/2 \rfloor\}$, the locus where all odd-power Galois-ring traces vanish to prescribed dyadic precisions. A lower bound that reads this description (a parity-breaking sieve, a Selberg-type minorant on the Witt filtration, an explicit construction) is not blocked by Proposition 1.
- (3) **A phase-aware correlation:** an inequality among the $S_n(\chi)$ at frequency n – aggregate cancellation of the signed sum, not a bound on each modulus. The roadmap’s open estimate asks for exactly this, a factor 4ℓ of cancellation across high 2-power orders that no per-character absolute value gives [1].

Numerical check. An exact-integer verifier over the class dumps confirms both theorems at $\ell = 12, \dots, 24$ for both endpoints. In every one of the fifteen cases the exceptional set of Theorem B is *empty*, though the theorem permits up to 10% of classes at $\ell = 12$; the worst class sits 0.3–13% off the mean. The measured $\sum_g D(g)^2$ equals $1.00\times$ the Sato–Tate prediction to three digits from $\ell = 16$ on, so the true exceptional set is far smaller than either theorem certifies.

Prior art. The method is the function-field large-sieve second moment, unconditional because the Riemann hypothesis here is Weil’s theorem. Pointwise asymptotics for *every* class are classical while $\ell \leq n/2 - \log_2 \ell - O(1)$ (Hayes [2]; effective form Hsu [3]), failing exactly in the last $\log_2 \ell$ coefficients – the entire Lemire gap. The Sato–Tate variance is the $q \rightarrow \infty$ theorem of Keating–Rudnick [4]; our range keeps $q = 2$ fixed, the wrong limit. The explicit endpoint statement at $\ell = \lceil n/2 \rceil - 1$ and fixed $q = 2$ we did not find in print; treat it as folklore made explicit.

References

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